CHE374 — Economic Analysis and Decision Making

Course Notes

Bonnie Zhuang

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Contents

Week 1 - Introduction and Time Value of Money **3**

1.1 - Interest Rate . . . 3

1.2 - Simple and Compound Interest . . . 5

1.3 - Subperiod Compounding & Effective Interest Rates . . . 8

Week 2 - Cash-flow Analysis **11**

2.1 - Cash-flow Diagrams . . . 11

2.2 - Categories of Regular Cash-flows . . . 13

2.3 - Cash-flow Equivalence . . . 15

2.4 - Cash-flow Factors . . . 16

2.5 - Worked Examples . . . 21

Week 3 - Financial Instruments **25**

3.1 - Mortgages . . . 25

3.2 - Bonds . . . 27

3.3 - Bond Examples . . . 30

Week 4 - Risk, Reward and Arbitrage **31**

4.1 - Capital Asset Pricing Model . . . 31

4.2 - Arbitrage . . . 35

Week 5 - Comparison Methods **40**

5.1 - Project Relationships and Alternatives . . . 40

5.2 - Present Worth and Annual Worth . . . 42

5.3 - Evaluating Mutually Exclusive Projects (PW/AW) . . . 44

Week 6 - Comparison Methods (Part 2) **47**

6.1 - Internal Rate of Return . . . 47

6.2 - Comparing Mutually Exclusive Projects Using IRR . . . 50

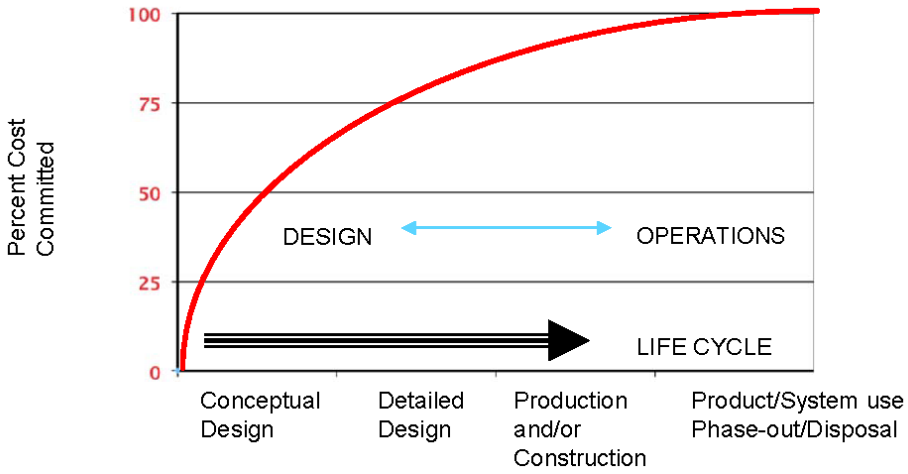
6.3 - Payback Period . . . 52

Week 1 - Introduction and Time Value of Money

1.1 - Interest Rate

Engineering Project Life Cycles

- All engineering projects have a life cycle:
  - Conceptualization → detailed economic analysis → go / no go decision → design → build → utilize / produce → phase out / dispose
- Most of the project's cost is **committed early**, during conceptual and detailed design, even though the money is mostly **spent later** (production, operation)
- We **invest** in projects to gain benefits (economic and societal)



Most cost is committed early, during design, before it is actually spent

Time Value of Money

- Borrowing (renting) money has a cost for the **borrower**, thus lending money *should* create value for the lender (investor/creditor)
- Costs and benefits occur in different time periods
- The cost/benefit today to the future is compared by **interest**

Example

You borrow \$10 from a lender today and return it a year later.

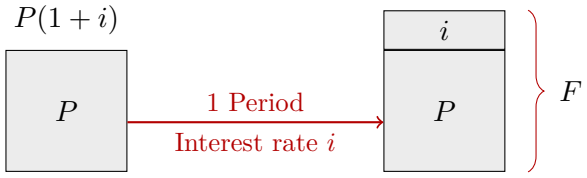
- You must return the \$10 borrowed ⇒ the **principal**
- The lender charges an extra amount, e.g. \$1 ⇒ the **interest**
- The **interest rate** is interest/principal:

$$i = \frac{1}{10} = 10\%$$

Present and Future Worth

$P =$ present worth/**Principal** $\quad F =$ future worth $\quad i =$ interest rate

$$F = P + Pi \qquad F = P(1+i)$$



- A **higher** payoff (F) is expected as i increases
- For a given F with increased i, P **decreases**

Interest Rate

- Expressed as %
- Vary widely over time and different types of investments.
  - A **T-Bill** (Treasury Bill, short term loan to the Bank of Canada) will earn the **lowest** rate of return, as it's considered "risk-free"
  - Canadian and provincial bonds pay typically **slightly higher** interest rates
  - Personal/small business loans are much riskier and earn **significantly higher** interest rates

Factors that Affect Interest Rates

- **Inflation**: prices generally tend to increase over time. Higher expected inflation generally leads to **higher interest rates**
- **Credit (default) Risk**: associated with the borrower being able to pay back interest and principal. Higher risk of default ⇒ **higher interest rates**
- **Liquidity Risk**: associated with being able to access the invested funds during the investment. The less liquid the investment (greater the risk), the **higher the interest rate**
- **Maturity Risk**: associated with the time horizon. Longer the maturity, longer payoff, increases credit (and potential liquidity) risk ⇒ **higher interest rates**

Generally, as **risk** increases, the **interest rate** increases.

Example

What would the impact be expected on the interest rate associated with a large new mining project under the following scenarios, and how would the current value be impacted?

- The expected time to complete the project construction increases
  Interest rate will increase as the maturity risk is directly incrased, raising risks of liquidity and credit. The present value would then decrease, from both the interest rate increasing and the time horizon increasing.
- The estimate amount of ore available to be mined increases
  The future cash flow would increase, thus the value of the project will increase, decreasing risk. Credit risk will drop but not others.

Summary

- **Interest** is money earned by investors for allowing others to use their money
- The **interest rate** is a rate at which interest is earned.
- The interest rate is determined by **risks** associated with the investment.

1.2 - Simple and Compound Interest

Present and Future Worth

$$P = \text{ amount of money today (principal)} \qquad F_N = \text{ future amount in year } N$$

$$i = \text{ interest rate} \qquad I = \text{interest amount}$$

$$F = P + I \qquad F_1 = P + Pi$$

Example GIC

You deposit \$5,000 in the bank for 1 year as a **Guaranteed Investment Certificate (GIC)**, i.e. you *lent* the bank the money for a fixed time period. After 1 year the bank pays 10% interest. You get $I = 5000(0.10) = \$500$ as "rent" for lending the \$5,000, and the \$5,000 is also returned.

- This is a **simple** interest loan: you receive the money when the GIC **matures**
- **Matures**: the borrower must pay the loan back

Simple vs Compound Interest

- **Simple** applies **only to the original principal** (GIC, aka Certificate of Deposit (CD) in the U.S.)
- **Compound** interest applies to the **principal AND all interest already accrued** (mortages, credit card debt...)
- Assume **compound** unless stated otherwise

Simple Interest

Beginning of Period	Amount Lent	Interest Amount	Amount Owed at Period End
1	P	Pi	$P + Pi$
2	$P + Pi$	Pi	$P + 2Pi$
3	$P + 2Pi$	Pi	$P + 3Pi$
...	...	...	...
N	$P + (N-1)Pi$	Pi	$P + NPi = P(1 + Ni)$

$$F_N = P(1 + Ni)$$

F_N : amount owed after N periods
P : principal
i : interest rate per period
N : number of periods

Compound Interest

Beginning of Period	Amount Lent	Interest Amount	Amount Owed at Period End
1	P	Pi	$P(1 + i)$
2	$P(1 + i)$	$P(1 + i)i$	$P(1 + i)^2$
3	$P(1 + i)^2$	$P(1 + i)^2 i$	$P(1 + i)^3$
...	...	...	...
N	$P(1 + i)^{N-1}$	$P(1 + i)^{N-1} i$	$P(1 + i)^N$

$$F_N = P(1 + i)^N$$

F_N : amount owed after N periods
P : principal
i : interest rate per period
N : number of periods

Example Simple Interest

If you borrow \$100 for 3 years at 10% annual interest (simple interest), how much interest will you pay, and how much will you owe after 3 years?

$$P = 100 \qquad i = 10\% \qquad N = 3$$

$$I = NPi = 3(100)(0.10) = \$30$$
$$F_3 = P + NPi = 100 + 30 = \$130$$

Example Compound Interest

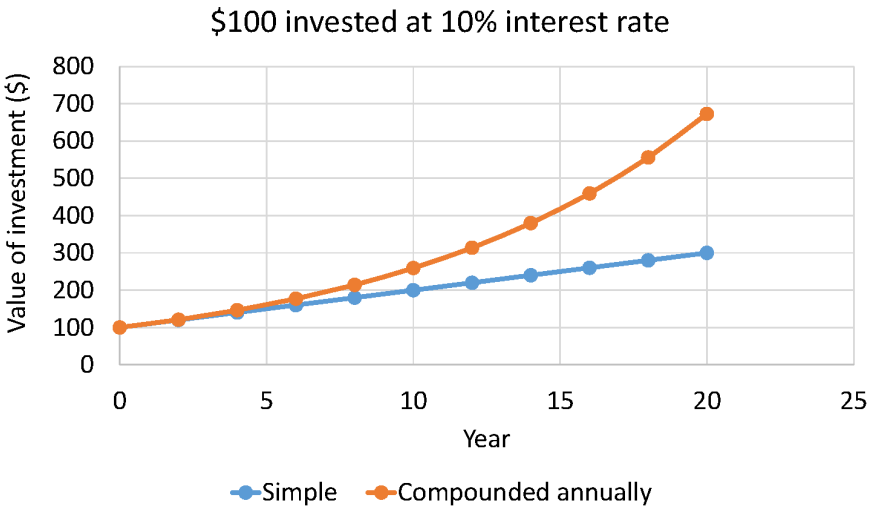
If you borrow \$100 for 3 years at 10% annual interest (compound interest), how much interest will

you pay, and how much will you owe after 3 years?

$$F_3 = P(1+i)^N = 100(1.10)^3 = \$133.10$$
$$I = F_3 - P = 133.10 - 100 = \$33.10$$

The extra \$3.10 compared to simple interest is interest earned on interest.

Simple Interest is Linear, Compound Interest is Exponential



\$100 invested at 10%: \$300 (simple) vs. \$673 (compound) after 20 years

Example The Power of Compounding

The Dutch West India Company (supposedly) paid \$24 to purchase Manhattan Island in 1626. If the company had invested the \$24 in a savings account that earned 7% interest per year, how much would it be worth today (2020, $N = 394$)?

$$\text{Simple: } F = 24 + 394(24)(0.07) \approx \$686$$
$$\text{Compound: } F = 24(1.07)^{394} \approx \$9{,}066{,}082{,}143{,}625$$

Year	Simple (7%)	Compound (7%)
1626	\$24	\$24
1676	\$108	\$707
1726	\$192	\$20,825
1776	\$276	\$613,448
1826	\$360	\$18,070,359
1876	\$444	\$532,299,016
1926	\$528	\$15,679,945,443
1976	\$612	\$461,884,545,916
2020	\$686	\$9,066,082,143,625

Summary

- Simple Interest: $F_N = P + NPi$
- Compound Interest: $F_N = P(1+i)^N$
- Remember: **interest rate** = interest/principal

1.3 - Subperiod Compounding & Effective Interest Rates

Compound Interest

- Recall: **compound** interest applies to the principal AND to all interest already accrued
- "To compound" (verb) is to reinvest the principal PLUS the interest and earn interest on both:

$$F = P(1+i)^N$$

Compound Multiple Times per Year

- r: **nominal interest rate** (usually for 1 year)
- m: times compounded in a year, aka **subperiods**
- i_s: **subperiod interest rate**

$$i_s = \frac{r}{m}$$

i_s : subperiod interest rate
r : nominal interest rate (usually per year)
m : number of subperiods per year

Example

You lend \$100. The nominal interest rate is 6%.
What is the total amount in the account at the end of the year if the compounding period is

$$P = 100 \qquad r = 6\%$$

a) 1 year:

$$\begin{aligned} F &= P(1+r) \\ &= 100(1+0.06) \\ &= 106 \end{aligned}$$

b) 3 months:

$$m = 4 \quad \Rightarrow \quad i_s = \frac{r}{m} = \frac{0.06}{4} = 0.015$$

$$\begin{aligned} F &= P(1+i_s)^m \\ &= 100(1+0.015)^4 \\ &= 106.14 \end{aligned}$$

Example Reflection

True or false: for the same nominal interest rate, the more frequently you compound, the more you earn at the end of the year.

True. Each extra compounding lets earlier interest start earning interest sooner ($106 vs. $106.14 above). The gain shrinks as m grows, though, approaching a finite limit (continuous compounding).

Effective Interest Rate

- How do we compare investments with different compounding periods?
- i_e: **effective interest rate**, the equivalent interest rate of an account that is compounded just **once** over the stated time period (usually 1 year)
- Set one year of subperiod compounding equal to one compounding at i_e:

$$P(1+i_e) = P\left(1+\frac{r}{m}\right)^m$$

$$\boxed{i_e = \left(1+\frac{r}{m}\right)^m - 1}$$

i_e : effective interest rate (per year)
r : nominal interest rate (per year)
m : number of subperiods per year

Example

Which account earns you more money after 1 year?

- Option A: $r = 6\%$, compounded monthly
- Option B: $r = 6.3\%$, compounded once a year

$$\text{A: } i_e = \left(1+\frac{0.06}{12}\right)^{12} - 1 = (1.005)^{12} - 1 = 6.17\%$$
$$\text{B: } i_e = r = 6.3\% \quad (m = 1)$$

Option B earns more, since $6.3\% > 6.17\%$. Compare the **effective** rates, not the nominal ones.

Continuous Compounding

- As the compounding period shortens (m increases), i_e **increases**
- As the compounding period becomes infinitesimally small, i_e approaches a **finite limit** $\Rightarrow$ **continuous compounding**

$$i_e = \lim_{m\to\infty}\left(1+\frac{r}{m}\right)^m - 1 \quad \Rightarrow \quad \boxed{i_e = e^r - 1}$$

- e.g. $r = 6\%$: $i_e = e^{0.06} - 1 = 6.18\%$, only slightly more than monthly (6.17%)

Summary

- Know the difference between **nominal** and **effective** interest rates, and how to convert between them
    1. Start with the nominal interest rate r for a stated time period (usually 1 year)
    2. Compound multiple (m) times within that time period, at $i_s = r/m$
    3. Effective interest rate: $i_e = (1 + r/m)^m - 1$
    4. Continuous compounding: $i_e = e^r - 1$

Week 2 - Cash-flow Analysis

2.1 - Cash-flow Diagrams

Remark (Project Life-Cycle Cost Categories).

- **Start**
  - **First (capital) cost**: expense to build/buy and install
- **During**
  - **Revenues (sales)**: receipts from sale of products/services
  - **O&M costs**: expenses incurred on a regular basis (electricity, labour, repairs)
  - **Overhaul**: a major capital expenditure partway through the asset's life
- **End**
  - **Salvage value**: net receipt from sale/disposal at the end of the asset's **useful** life
  - **Scrap value**: value of the materials at the end of the asset's **physical** life
  - **Disposal costs**: cost to dispose of waste (removal, shipping, tipping fees, etc.)

Definition - Cash-flow Table

A table showing the "money consequences" of a situation and the timing of the cash-flows: for each point in time, the activity and the resulting cash-flow. Costs/benefits occurring at the start, during, or end of a project are together called the **project life-cycle costs**.

Example **Car Ownership Cash-flow Table**

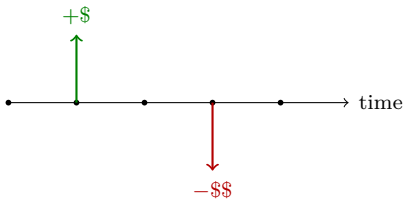
Year	Cash-flow	Activity
Start of Year 1	−\$24,500	Car purchased now
End of Year 1	−\$1,500	Service cost
End of Year 2	−\$1,500	Service cost
End of Year 3	−\$1,500	Service cost
End of Year 4	−\$1,500, +\$5,000	Service cost & car sold for \$5,000

The purchase is the first cost (at $t = 0$), service is an O&M cost, and the \$5,000 sale is the salvage value.

Definition - Cash-flow Diagram

A simple graph summarizing the **timing** and **magnitude** of cash-flows.

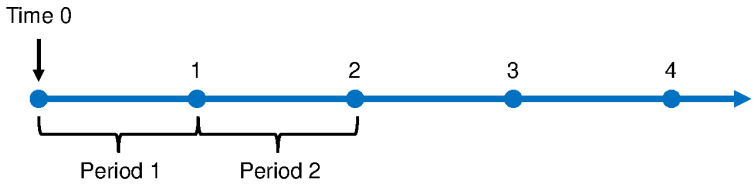
- x-axis: discrete time periods
- (implicit) y-axis: the size and direction of each cash-flow, drawn as a vertical arrow
  - **Up** arrow: a receipt (inflow, +\$)
  - **Down** arrow: a disbursement (outflow, −\$)



Up = inflow (receipt); down = outflow (disbursement)

Remark (Diagram Conventions).

- The end of one period is the beginning of the next: time 0 is the start of period 1, time 1 is the end of period 1 and the start of period 2, and N periods from now is the end of period N / start of period $N+1$.
- Unless stated otherwise: interest compounds once per period, and cash-flows occur at the **end** of the period.
- Cash-flows occurring *during* a period are **summed** and accounted for at the **end** of that period.
- First costs are typically accounted for at **time 0**.



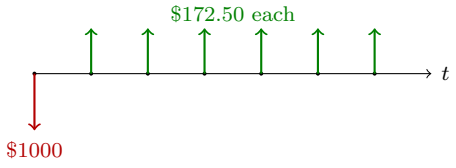
Example Loan Repayment Diagram

$1,000 is borrowed on January 1 and repaid in 6 monthly installments of $172.50 ($i = 1\%$ per month). The time line's origin is the start of the analysis (time 0 = January 1, intervals are monthly).

- **Borrower's** perspective: an inflow of $1,000 at $t = 0$, and outflows of $172.50 at $t = 1, \ldots, 6$
- **Bank's** perspective (the diagram flips): an outflow of $1,000 at $t = 0$, and inflows of $172.50 at $t = 1, \ldots, 6$



Borrower



Bank (flipped)

Example Reflection: Young Professional

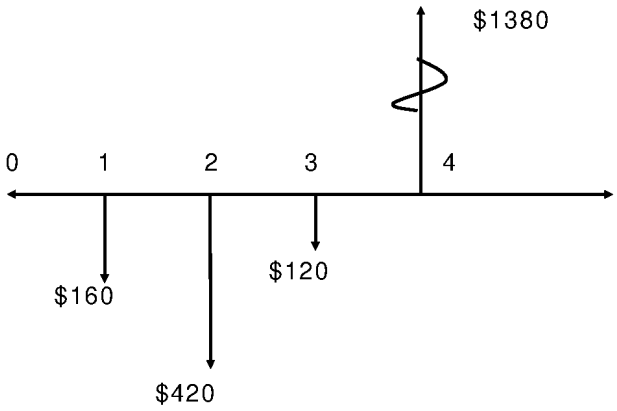
Construct a cash-flow diagram for one month (assume 4 weeks per month):

- Monthly income: $2,200 (end of each month)

- Rent incl. utilities: \$700 (end of each month)
- Weekly food and entertainment: \$120
- Telephone bill: \$40 (end of the first week of each month)
- Credit card purchases: \$300 (end of the second week of each month)

Use weeks as periods and net everything at the end of each week:

$$\text{Week } 1: -120 - 40 = -\$160$$
$$\text{Week } 2: -120 - 300 = -\$420$$
$$\text{Week } 3: -\$120$$
$$\text{Week } 4: 2200 - 700 - 120 = +\$1{,}380$$



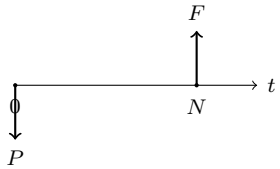
2.2 - Categories of Regular Cash-flows

Five Types of (Regular) Cash-flows

1. Single payments or receipts
2. Perpetuity
3. Annuity
4. Arithmetic gradient
5. Geometric gradient

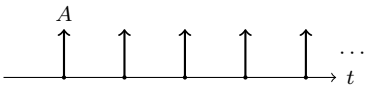
Definition - Single Payment/Receipt

A single cash-flow P (now) or F (in the future), occurring only **once**, N periods apart.



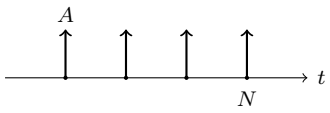
Definition - Perpetuity

A constant cash-flow A occurring every period, **forever** (an *infinite annuity*).



Definition - Annuity

A constant (uniform) cash-flow A occurring every period, for a **finite** number of periods N.

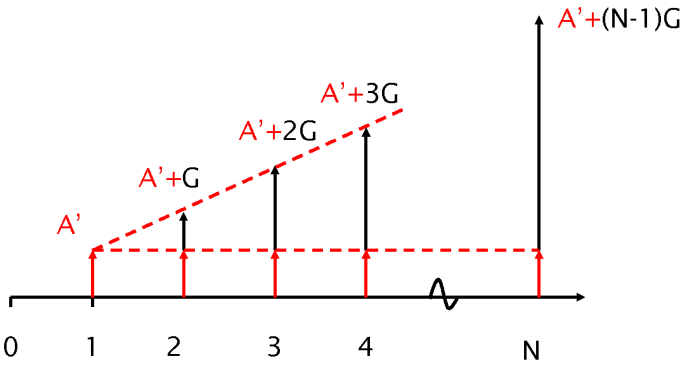


Definition - Arithmetic Gradient

A cash-flow that grows (or shrinks) by the same **amount** G each period, starting from a base A' at $t = 1$:

$$A',\ A' + G,\ A' + 2G,\ \ldots,\ A' + (N-1)G.$$

It splits into an annuity of A' (red, below) plus a pure gradient $G, 2G, \ldots$ on top (black).

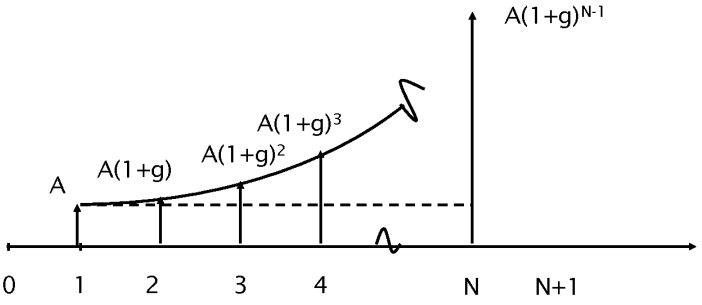


Definition - Geometric Gradient

A cash-flow that grows (or shrinks) by the same **percentage** g each period, starting from A at $t = 1$:

$$A,\ A(1+g),\ A(1+g)^2,\ \ldots,\ A(1+g)^{N-1}.$$

There **must** be a cash-flow at the end of period 1, otherwise there is nothing for the percentage to grow from.



2.3 - Cash-flow Equivalence

Three Concepts of Equivalence

- **Mathematical**: consequence of the mathematical relationship between time and money
- **Market**: consequence of the ability to exchange one cash-flow for another at zero cost
- **Decisional**: due to indifference on the part of the decision maker among available choices

Definition - Mathematical Equivalence

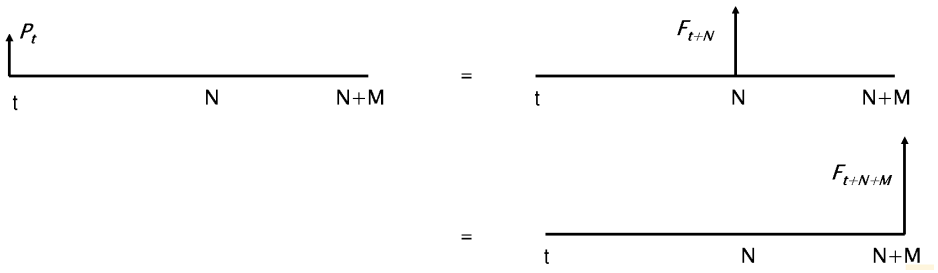
A consequence of the mathematical relationship between time and money. Two cash-flows P_t (at time t) and F_{t+N} (at time $t+N$) are equivalent at rate i if

$$F_{t+N} = P_t(1+i)^N \iff P_t = \frac{F_{t+N}}{(1+i)^N}.$$

If a further cash-flow F_{t+N+M} is also equivalent to P_t, then

$$F_{t+N+M} = P_t(1+i)^{N+M} = F_{t+N}(1+i)^M \iff P_t = \frac{F_{t+N+M}}{(1+i)^{N+M}},$$

i.e. equivalent amounts compound/discount consistently with each other.



Definition - Market Equivalence

A consequence of being able to exchange one cash-flow for another at **zero cost**.

- **Borrowing**: exchanging a future cash-flow for a present one
- **Lending/investing**: giving up a present cash-flow for a future one

In real markets, borrowing and lending rates usually **differ**.

Example T-bill Market Equivalence

A T-bill is a short-term government debt obligation (maturity ≤ 1 year), usually sold in \$1,000 denominations. A \$1,000 T-bill maturing in 6 months currently sells for \$990.10: some holders are willing to sell at this price and others to buy. The market has agreed this corresponds to an interest rate of 1% over 6 months. Confirm:

$$i = \frac{1000}{990.10} - 1 = 0.0100 = 1\%$$

Definition - Decisional Equivalence

Two cash-flows P_t and F_{t+N} are equivalent for a decision maker if they are **indifferent** between them. Here the interest rate is not known a priori — instead, an **implied** rate can be backed out from the decision that the two are equivalent.

Example Implied Interest Rate

A manufacturer needs 1,000 kg of aluminium on August 15 but is offered 1,100 kg on August 22 instead (same price, paid at the end of August).

$$i = \frac{1100}{1000} - 1 = 10\% \text{ for one week}$$

The real question is whether the extra 100 kg is worth the cost of the one-week delay to this decision maker. If they are indifferent, 10%/week is their implied rate.

Remark. Financial analysis uses **mathematical** and **market** equivalence to decide whether to invest or not invest.

2.4 - Cash-flow Factors

Remark (Standing Assumptions). Unless otherwise noted:

1. Interest compounds once per period.
2. Cash-flow occurs at the end of the period.
3. Time 0 is period 0, i.e. the start of period 1.
4. All periods are the same length.

Definition - Cash-flow Symbols

- ***P***: a single cash-flow today
- ***F***: a single cash-flow in the future
- ***A***: a finite series of equal cash-flows at regular intervals (annuity)
- ***G***: a cash-flow growing by a fixed **amount** each period (arithmetic gradient) or fixed **percentage** g each period (geometric gradient, "Geom")

Definition - Factor Notation

$(X/Y, i, N)$ converts a cash-flow of type Y into the equivalent amount of type X at rate i over N periods: $X = Y \cdot (X/Y, i, N)$. E.g. $F = P(F/P, i, N)$; for geometric gradients, $P = G(P/\text{Geom}, i, g, N)$.

Factor	Name	Given	Find
$(F/P, i, N)$	Compound amount factor	P	F
$(P/F, i, N)$	Present worth factor	F	P
$(A/F, i, N)$	Sinking fund factor	F	A
$(F/A, i, N)$	Series compound amount factor	A	F
$(A/P, i, N)$	Capital recovery factor	P	A
$(P/A, i, N)$	Series present worth factor	A	P
$(P/G, i, N)$	Arithmetic gradient to present worth	G	P
$(P/\text{Geom}, i, g, N)$	Geometric gradient to present worth	A, g	P

Remark (Relating Factors). Factors invert: since $P = A(P/A, i, N)$,

$$A = \frac{P}{(P/A, i, N)} = P(A/P, i, N) \quad \Rightarrow \quad (A/P, i, N) = \frac{1}{(P/A, i, N)}.$$

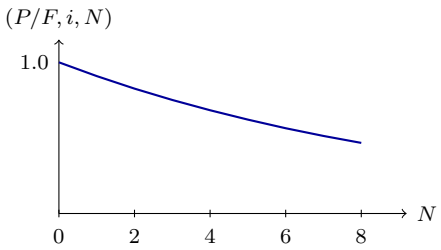
The same trick relates any pair of factors.

Definition - Single-Payment Factors

P and F are related by compounding one period at a time, $F_n = F_{n-1}(1+i)$, so $F = P(1+i)^N$:

$$(F/P, i, N) = (1+i)^N, \qquad (P/F, i, N) = \frac{1}{(1+i)^N}$$

P : single cash-flow at time 0
F : single cash-flow at time N
i : interest rate per period
N : number of periods



$(1+i)^{-N}$ *decays toward* 0 *as* N *grows (shown for* $i = 10\%$*)*

Example F/P

If you had \$2,000 now and invested it at 10%, how much would it be worth in 8 years?

$$F = 2000(F/P, 10\%, 8) = 2000(1.10)^8 = 2000(2.1436) = \$4{,}287.18$$

Example P/F: Solving for i

You invest \$1,100 today and expect \$2,000 in 8 years. What is the interest rate?

$$1100 = 2000(P/F, i, 8) \;\Rightarrow\; (1+i)^8 = \frac{2000}{1100} = 1.8182 \;\Rightarrow\; i = 1.8182^{1/8} - 1 = 7.76\%$$

Example Present Value of a Perpetuity

Invest \$100 and get paid 5% interest in cash every year, forever. Each year the \$5 of interest is paid out, so the balance never changes:

Year	Interest	Cash-flow	Amount in bank at year end
0	—	−100	100
1	5	+5	100
2	5	+5	100
3	5	+5	100
⋮	⋮	⋮	⋮

So \$100 today is equivalent to \$5/year forever, i.e. $P = A/i = 5/0.05 = 100$. In general, a constant amount A paid forever has a **finite** present value:

$$P = \frac{A}{1+i} + \frac{A}{(1+i)^2} + \cdots = \frac{A}{i}$$

$$P = \frac{A}{i}$$

P : present worth, one period before the first payment
A : payment every period, forever
i : interest rate per period

Definition - Present Worth of an Annuity

An annuity of A for N periods is a perpetuity starting now minus a perpetuity starting at N (worth A/i at time N, discounted back):

$$(P/A, i, N) = \frac{1}{i} - \frac{1}{i(1+i)^N} = \frac{(1+i)^N - 1}{i(1+i)^N}$$

Combining with (F/P) and inverting gives the rest of the uniform-series factors:

$$(P/A, i, N) = \frac{(1+i)^N - 1}{i(1+i)^N}, \qquad (A/P, i, N) = \frac{i(1+i)^N}{(1+i)^N - 1}$$

$$(F/A, i, N) = \frac{(1+i)^N - 1}{i}, \qquad (A/F, i, N) = \frac{i}{(1+i)^N - 1}$$

A : equal payment at the end of each period $1, \ldots, N$
P : present worth at time 0
F : future worth at time N (same time as the last payment)
i : interest rate per period
N : number of payments

Definition - Present Worth of an Arithmetic Gradient

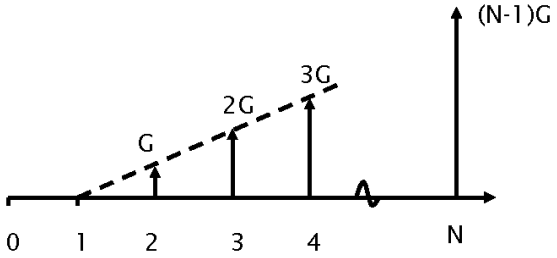
For a gradient contributing $G, 2G, \ldots, (N-1)G$ starting at $t = 2$ (**no cash-flow at $t = 0$ or $t = 1$**):

$$(P/G, i, N) = \frac{1}{i^2}\left(1 - \frac{1 + iN}{(1+i)^N}\right)$$

A series with initial amount A growing by G each period splits into an annuity plus a gradient:

$$P = A(P/A, i, N) + G(P/G, i, N)$$

A : cash-flow at $t = 1$
G : amount added each period after
i : interest rate per period
N : number of periods



The pure gradient: nothing at $t = 0, 1$; G at $t = 2, \ldots, (N-1)G$ at $t = N$

Remark (Sign Combinations for Arithmetic Gradients). Besides $G = 0$, four cases:

- $A > 0$, $G > 0$: positive, increasing
- $A > 0$, $G < 0$: positive, decreasing

- $A < 0$, $G > 0$: negative, becoming less so
- $A < 0$, $G < 0$: negative, becoming more so

Definition - Present Worth of a Geometric Gradient

For a series A, $A(1+g)$, $A(1+g)^2, \ldots$ starting at $t = 1$:

$$(P/\text{Geom}, i, g, N) = \frac{1 - \left(\frac{1+g}{1+i}\right)^N}{i - g} = \frac{(P/A, i_o, N)}{1+g}, \qquad i_o = \frac{1+i}{1+g} - 1$$

g : growth rate per period
i : interest rate per period
i_o : growth-adjusted interest rate
N : number of periods

Remark. Regardless of the pattern, present value always follows from **discounting each cash-flow individually and summing**. Each period a cash-flow is moved forward, **multiply** by $(1+i)$; each period back, **divide** by $(1+i)$:

$$P = \frac{F_1}{1+i} + \frac{F_2}{(1+i)^2} + \cdots = \sum_t \frac{F_t}{(1+i)^t}$$

F_t : net cash-flow at time t
i : interest rate per period

Factor Table ($i = 10\%$)

i = 10% — Discrete Compounding, Discrete Cash Flows

	SINGLE PAYMENT		UNIFORM SERIES				Arithmetic Gradient
	Compound Amount Factor	Present Worth Factor	Sinking Fund Factor	Uniform Series Factor	Capital Recovery Factor	Series Present Worth Factor	Series Factor
N	(F/P,i,N)	(P/F,i,N)	(A/F,i,N)	(F/A,i,N)	(A/P,i,N)	(P/A,i,N)	(A/G,i,N)
1	1.1000	0.90909	1.0000	1.0000	1.1000	0.90909	0.00000
2	1.2100	0.82645	0.47619	2.1000	0.57619	1.7355	0.47619
3	1.3310	0.75131	0.30211	3.3100	0.40211	2.4869	0.93656
4	1.4641	0.68301	0.21547	4.6410	0.31547	3.1699	1.3812
5	1.6105	0.62092	0.16380	6.1051	0.26380	3.7908	1.8101
6	1.7716	0.56447	0.12961	7.7156	0.22961	4.3553	2.2236
7	1.9487	0.51316	0.10541	9.4872	0.20541	4.8684	2.6216
8	2.1436	0.46651	0.08744	11.436	0.18744	5.3349	3.0045
9	2.3579	0.42410	0.07364	13.579	0.17364	5.7590	3.3724
10	2.5937	0.38554	0.06275	15.937	0.16275	6.1446	3.7255
11	2.8531	0.35049	0.05396	18.531	0.15396	6.4951	4.0641
12	3.1384	0.31863	0.04676	21.384	0.14676	6.8137	4.3884
13	3.4523	0.28966	0.04078	24.523	0.14078	7.1034	4.6988
14	3.7975	0.26333	0.03575	27.975	0.13575	7.3667	4.9955
15	4.1772	0.23939	0.03147	31.772	0.13147	7.6061	5.2789
16	4.5950	0.21763	0.02782	35.950	0.12782	7.8237	5.5493
17	5.0545	0.19784	0.02466	40.545	0.12466	8.0216	5.8071
18	5.5599	0.17986	0.02193	45.599	0.12193	8.2014	6.0526
19	6.1159	0.16351	0.01955	51.159	0.11955	8.3649	6.2861
20	6.7275	0.14864	0.01746	57.275	0.11746	8.5136	6.5081

Example Reflection: Lump Sums vs. Annuity

Which option do you prefer if interest is 8%?

a) $20,000 now and $50,000 at the end of 10 years
b) Ten equal annual receipts of $7,000

$$P_a = 20{,}000 + 50{,}000(P/F, 8\%, 10) = 20{,}000 + 50{,}000(0.46319) = \$43{,}160$$
$$P_b = 7{,}000(P/A, 8\%, 10) = 7{,}000(6.7101) = \$46{,}971$$

Prefer (b): it has the higher present value.

Example Reflection: Annuity at a Lower Rate

$1/year for 5 years has a present value of $3.7908 at 10%. Which is a likely value at 9%? (a) 3.8897 (b) 5.0000 (c) 3.7908 (d) 2.3262
(a) 3.8897. A lower rate discounts less, so PV rises above 3.7908, but it must stay below the undiscounted total of $5. Check: $(P/A, 9\%, 5) = 3.8897$.

2.5 - Worked Examples

Example Investment Decision

An investor can buy a tract of land that will be worth $10,000 in six years. If the rate of return on land is 8%, how much should the investor be willing to pay now?

$$P = 10{,}000(P/F, 8\%, 6) = \frac{10{,}000}{(1.08)^6} = \$6{,}301.70$$

Example The Magic of Compounding

At age 20 you save and invest $1.00 each day ($365/year) until age 80, at 10% annual interest.

$$F = 365(F/A, 10\%, 60) = 365 \cdot \frac{(1.10)^{60} - 1}{0.10} \approx \$1{,}107{,}708$$

You become a millionaire.

Example F and P with Subperiod Compounding

How much is accumulated over 5 years in each savings plan?

- $2,000 today at 10% compounded semi-annually: $i_s = 5\%$, $N = 10$

$$F = 2000(1.05)^{10} = \$3{,}257.79$$

- $1,000 today at 12% compounded monthly: $i_s = 1\%$, $N = 60$

$$F = 1000(1.01)^{60} = \$1{,}816.70$$

How much must you invest today to get \$2,000 in 5 years at 8% compounded quarterly? $i_s = 2\%$, $N = 20$:

$$P = 2000(P/F, 2\%, 20) = \frac{2000}{(1.02)^{20}} = \$1{,}345.94$$

Example Mortgage/Amortization

A \$300,000 mortgage at a nominal 9% (mortgages are quoted with semi-annual compounding) is amortized over 15 years. What is the monthly payment?
Semi-annual rate = 4.5%. The equivalent monthly rate satisfies $(1 + i_m)^6 = 1.045$:

$$i_m = 1.045^{1/6} - 1 = 0.7363\%, \qquad N = 15 \times 12 = 180$$

$$A = 300{,}000(A/P, 0.7363\%, 180) = \$3{,}013.56 \text{ per month}$$

That is about \$36,163/year, over 70% of a \$50k starting salary (before tax), so it doesn't fit the budget.

Example Sinking Fund

A "targeted" F calculation: save up to pay a future expenditure (e.g. replacing a machine). A roof must be replaced in 10 years for an estimated \$28,000. What equal end-of-year deposits at 8% are needed?

$$A = 28{,}000(A/F, 8\%, 10) = 28{,}000 \cdot \frac{0.08}{(1.08)^{10} - 1} = \$1{,}932.83 \text{ per year}$$

Example Retirement Planning

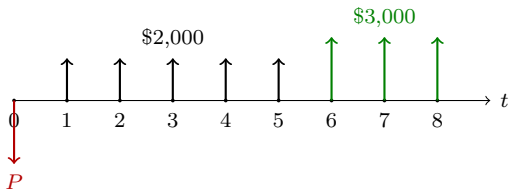
You are 20 and want \$1,000,000 at retirement at 65 (45 years). What equal end-of-year savings are needed at 7%? At 10%?

$$A_{7\%} = 1{,}000{,}000(A/F, 7\%, 45) = \$3{,}499.57 \text{ per year}$$
$$A_{10\%} = 1{,}000{,}000(A/F, 10\%, 45) = \$1{,}391.00 \text{ per year}$$

Example Uneven Annuity Segments

Claudia wants to deposit P now so that she can withdraw \$2,000/year for 5 years and then \$3,000/year for the next 3 years, at 8%. Discount the second annuity back to $t = 5$, then to $t = 0$:

$$\begin{aligned} P &= 2{,}000(P/A, 8\%, 5) + 3{,}000(P/A, 8\%, 3)(P/F, 8\%, 5) \\ &= 2{,}000(3.9927) + 3{,}000(2.5771)(0.68058) \\ &= 7{,}985.42 + 5{,}261.79 = \$13{,}247.21 \end{aligned}$$



Example Present Value with an Arithmetic Gradient: Investment

An investment of \$400,000 pays benefits of 20% (\$80,000) in year 1, with an additional \$15,000 added to each previous year's benefit, for 5 years. At a 10% discount rate, is it worth it?

$$\begin{aligned} P &= -400{,}000 + 80{,}000(P/A, 10\%, 5) + 15{,}000(P/G, 10\%, 5) \\ &= -400{,}000 + 80{,}000(3.7908) + 15{,}000(6.8618) \\ &= -400{,}000 + 406{,}190 = +\$6{,}190 \end{aligned}$$

Barely worth it: the benefits' PV just exceeds the \$400,000 cost.

Example Present Value with an Arithmetic Gradient: Maintenance

A machine costs \$155 to maintain this year, with maintenance costing \$35 more each subsequent year, over an 8-year life at 6%. How much must be invested now to cover maintenance, and what is the Equivalent Annual Cost (EAC)?

$$P = 155(P/A, 6\%, 8) + 35(P/G, 6\%, 8) = 155(6.2098) + 35(19.8416) = \$1{,}656.97$$
$$\text{EAC} = P(A/P, 6\%, 8) = 1{,}656.97(0.16104) = \$266.83 \text{ per year}$$

With a \$1,000 overhaul at the start ($t = 0$):

$$\text{EAC} = (1{,}656.97 + 1{,}000)(A/P, 6\%, 8) = \$427.87 \text{ per year}$$

Example Future Value with Periodic Dividends

\$100,000 is loaned at 15% (interest compounded annually, not paid out), with a \$5,000 dividend at the end of every third year for 15 years, reinvested at 15%. How much do you have at the end?
Principal: $100{,}000(1.15)^{15} = \$813{,}706$. The dividends are an annuity with 3-year periods ($N = 5$), so use the 3-year effective rate $i_3 = (1.15)^3 - 1 = 52.09\%$:

$$\begin{aligned} F &= 100{,}000(F/P, 15\%, 15) + 5{,}000(F/A, 52.09\%, 5) \\ &= 813{,}706 + 68{,}510 = \$882{,}216 \end{aligned}$$

Example A Business Example

Over 3 years a small business has:

- sales of \$240k, \$360k, \$480k per year, received in equal monthly amounts (\$20k, \$30k, \$40k/month)
- investments of \$15k/quarter in year 1, \$20k/quarter in year 2, \$25k/quarter in year 3
- an initial investment of \$100k, and a salvage value of \$70k after 3 years

What is the PV at 12%, analysed on monthly cash-flows?
Monthly rate $i = 12\%/12 = 1\%$. Each year's sales are a 12-month annuity shifted to the start of

its year; quarterly investments fall at months $3, 6, \ldots, 36$:

$$\begin{aligned} P = & -100{,}000 + \left[20{,}000 + 30{,}000(P/F, 1\%, 12) + 40{,}000(P/F, 1\%, 24)\right](P/A, 1\%, 12) \\ & - \sum_{q=1}^{12} Q_q (P/F, 1\%, 3q) + 70{,}000(P/F, 1\%, 36) \\ \approx & \$633{,}463 \end{aligned}$$

where Q_q is \$15k, \$20k or \$25k depending on the year.

Week 3 - Financial Instruments

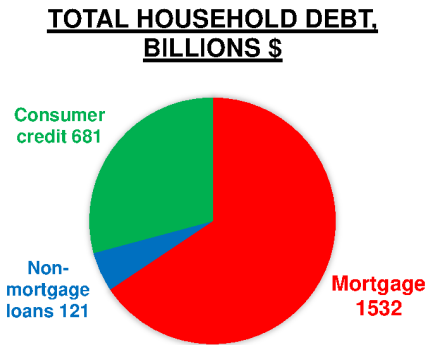
3.1 - Mortgages

Definition - Mortgage

A loan secured by **real property**: you borrow money and use the property as collateral. From French law, "death contract" (*mort-* = death): the pledge ends when the obligation is fulfilled.

Importance of Mortgages

- The most common way to finance a home: new purchases with a mortgage outnumber those without by a factor of 6 (CMHC)
- Mortgages are by far the largest share of Canadian household debt



Total household debt, billions $ (Statistics Canada)

Mortgage Terminology

Buying a $500,000 house:

- **Principal**: the amount you borrow, up to some fraction of the price (e.g. 80% ⇒ $400,000)
- **Down payment**: the rest, which you pay yourself ($100,000)
- **Loan-to-value ratio (LTV)**: mortgage loan / value of house = 400,000/500,000 = 0.8
- **Mortgage rate**: the interest rate the bank charges on the principal (e.g. 5%/year)
- **Amortization period**: the agreed time horizon for full repayment (e.g. 25 years)
- **Term**: the shorter period over which the agreed rate is fixed (e.g. 5 years). After the term ends, the rate is **renegotiated** and the payments may change

Remark.

- Payments are calculated based on the **amortization period**, **not** the term!
- Canadian mortgage rates are quoted as nominal annual rates with **semi-annual compounding**, so convert to the equivalent monthly rate before using $(A/P, i, N)$

$$(1+i_m)^6 = 1+\frac{r}{2} \quad \Rightarrow \quad i_m = \left(1+\frac{r}{2}\right)^{1/6} - 1$$

$$A = P(A/P, i_m, N)$$

r : quoted (nominal annual) mortgage rate
i_m : equivalent monthly rate
P : principal
N : amortization period in months
A : monthly payment

Example Mortgage Payments Across Terms

A \$500,000 home with a 20% down payment. The mortgage rate is 5%, with a 5-year term and a 25-year amortization period.

a) What is the monthly payment during this term?
Principal $P = 0.8(500{,}000) = \$400{,}000$. Semi-annual rate 2.5%, so the monthly rate satisfies $(1+i_m)^6 = 1.025$:

$$i_m = 1.025^{1/6} - 1 = 0.41239\%, \qquad N = 25 \times 12 = 300$$

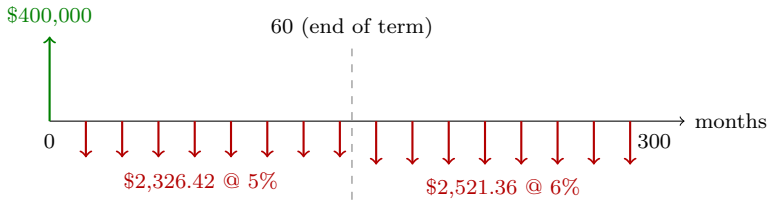
$$A = 400{,}000(A/P, 0.41239\%, 300) = \$2{,}326.42 \text{ per month}$$

b) After the term, the new mortgage rate is 6%. What is the new monthly payment?
The amount still owed after 60 payments is the PV of the 240 remaining payments at the old rate:

$$B_{60} = 2{,}326.42(P/A, 0.41239\%, 240) = \$354{,}030$$

Re-amortize this over the remaining 20 years at the new rate, $i'_m = 1.03^{1/6} - 1 = 0.49386\%$:

$$A' = 354{,}030(A/P, 0.49386\%, 240) = \$2{,}521.36 \text{ per month}$$



Borrower's view: payments step up at the renewal

Some Canadian Mortgage Rules

- Since 2012, amortization periods of up to 25 years are allowed
- For 25-year amortization, the down payment must be **at least 20%**

- For shorter amortization periods, the down payment may be smaller ($< 20\%$), but **mortgage insurance** is required
- Real-life mortgages can be more complicated: rates can be variable within a term, you can pay the mortgage down faster, ...

Summary

1. Mortgages are loans backed by real property (the most common way to finance a home)
2. Payments are calculated from the principal, mortgage rate, and **amortization period**
3. At the end of each term, the next payments are set by the **amount still owed**, the **new rate**, and the **remaining amortization period**

3.2 - Bonds

Definition - Bond

A financial instrument issued by firms and governments (federal, provincial, municipal) to raise funds for projects. It is a special form of loan, usually long-term (up to 30 years). The issuer (the borrower) promises to pay:

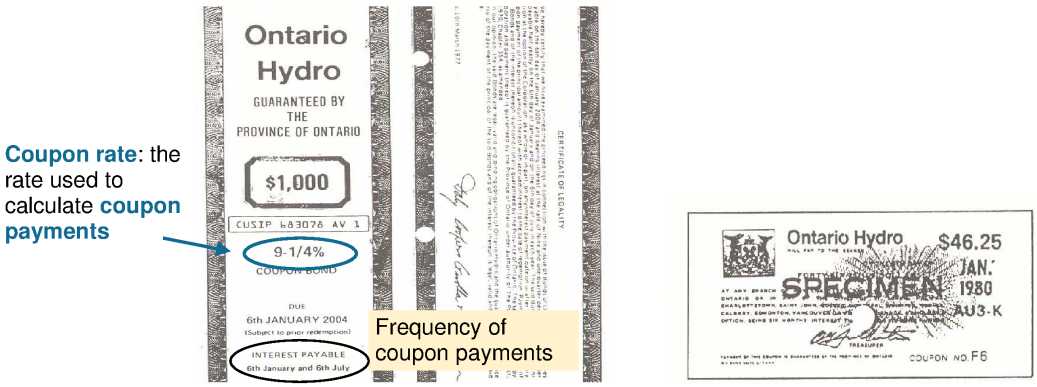
- a stated amount at specified intervals for a defined period: the **coupon payments**
- a final amount at a specific date: the **face value**

Bonds are usually **tradeable at any time**.

Reading a Bond (Ontario Hydro)

- **Face value**: the amount paid at maturity ($1,000)
- **Maturity date**: when the face value is paid (January 6, 2004)
- **Coupon rate**: the (nominal annual) rate used to calculate coupon payments ($9\frac{1}{4}\%$)
- **Coupon frequency**: how often coupons are paid (January 6 and July 6, i.e. semi-annually)

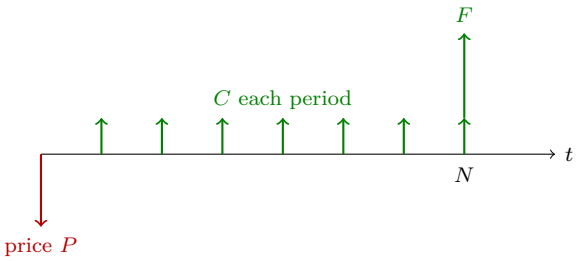




Each semi-annual coupon $= 1000 \times 9.25\%/2 = \$46.25$, matching the coupon shown.

Bond Cash-flow Diagram

From the bondholder's view: pay the price P now, receive coupon C every period for N periods, plus the face value F at maturity.



$$P = C(P/A, i, N) + F(P/F, i, N)$$

P : bond price
C : coupon payment per period ($= F \times$ coupon rate/coupons per year)
F : face value
i : yield per coupon period
N : number of coupon periods to maturity

Example How Much Is This Bond Worth?

Ontario Hydro will pay you \$1,000 in 10 years, plus coupons at 9.25% paid every 6 months. If the bank pays 10% interest, what amount must be deposited now to produce the same cash-flows?
$C = \$46.25$, $N = 20$ half-years, $i = 10\%/2 = 5\%$ per half-year:

$$\begin{aligned} P &= 46.25(P/A, 5\%, 20) + 1000(P/F, 5\%, 20) \\ &= 46.25(12.4622) + 1000(0.37689) = \$953.27 \end{aligned}$$

Would you be willing to pay \$953?
Only if the bond were as safe as the bank. Ontario Hydro might default, so you should demand a lower price.

Definition - Yield

Risk == reward: the buyer must be compensated for the risk of default by paying a **lower** price. The **yield** is the (hypothetical) interest rate that makes the discounted bond payments equal to the price. The yield reflects the time value of money for the bond's payments.

Example Yield from a Price

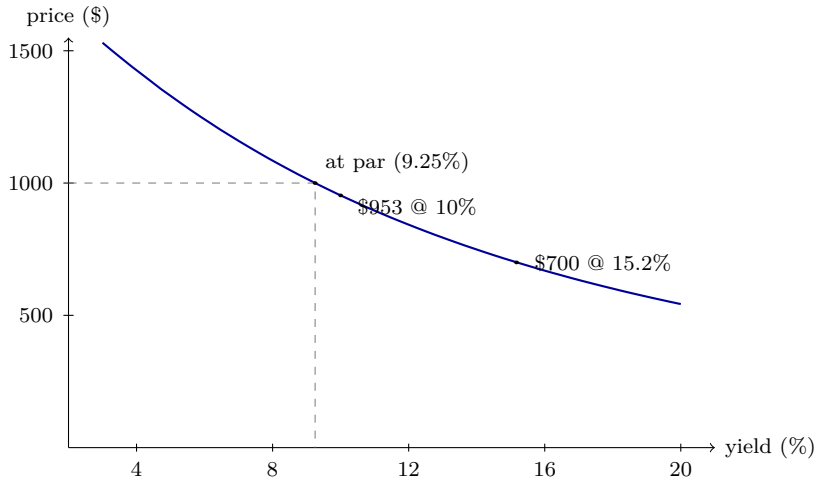
If the highest price you are willing to pay for the bond above is \$700, what annual interest rate gives that price?

$$700 = 46.25(P/A, i, 20) + 1000(P/F, i, 20)$$

Solving numerically (trial and error or a solver): $i = 7.587\%$ per half-year, i.e. a nominal yield of $\approx 15.2\%$/year (effective 15.75%).

Bond Price Is Determined by Yield

- Price = payments discounted at the yield, so a **higher yield** means a **lower price**
- Comparing the yield to the coupon rate:
  - Yield = coupon rate $\Rightarrow$ price = face value (**at par**)
  - Yield > coupon rate $\Rightarrow$ price < face value (**at a discount**)
  - Yield < coupon rate $\Rightarrow$ price > face value (**at a premium**)



Ontario Hydro bond ($1,000 face, 9.25% semi-annual coupon, 10 years)

Example Reflection

If the bond yield increases, does the price of the bond increase or decrease?
The price **decreases**: the same payments are discounted at a higher rate.

Yield Depends on Risk and Time to Maturity

Country	2-yr gov't yield (2019, %)
Argentina (1 yr)	32.59
Brazil	5.69
Canada	1.521
China	2.700
Germany	−0.828
Uganda	16.000
Ukraine	16.97
United Kingdom	0.449
United States	1.575

More stable economies ⇒ lower yield

Maturity (yrs)	Gov't of Canada yield (%)
3	1.00
5	1.50
7	2.25
10	2.25
Long (30)	2.75

Longer time to maturity ⇒ higher yield

This is the same idea as the Week 1 interest-rate factors: yield rises with **credit (default) risk** and **maturity risk**.

Summary

1. **Face value, coupon payments/rate, and maturity** are defined by the bond itself
2. The **yield** is determined by the market (how risky the investment is, interest rate expectations)
3. Bond prices are calculated by discounting the payments at the bond yield

3.3 - Bond Examples

Example Bond Price

A \$10,000 bond maturing in 8 years has a 12% coupon payable quarterly. If the yield is 10%, how much would you pay for the bond?
$C = 10{,}000 \times 12\%/4 = \$300$ per quarter, $N = 32$ quarters, $i = 10\%/4 = 2.5\%$ per quarter:

$$\begin{aligned} P &= 300(P/A, 2.5\%, 32) + 10{,}000(P/F, 2.5\%, 32) \\ &= 300(21.8492) + 10{,}000(0.45377) = \$11{,}092.46 \end{aligned}$$

Price > face value, because the coupon rate (12%) exceeds the yield (10%).

Example Yield on a Bond

You buy a bond for \$1,000. The issuer pays a \$45 coupon every 6 months and a face value of \$880 at the end of 2 years. What is the **effective annual** yield?
Solve for the semi-annual rate i with $N = 4$:

$$1000 = 45(P/A, i, 4) + 880(P/F, i, 4) \;\Rightarrow\; i = 1.570\% \text{ per half-year}$$

$$i_e = (1 + 0.01570)^2 - 1 = 3.16\%$$

The yield is low despite the 9%/year coupons, because you only get \$880 back on a \$1,000 purchase.

Week 4 - Risk, Reward and Arbitrage

4.1 - Capital Asset Pricing Model

Definition - Valuation

The analytical process of determining the current (or projected) worth of an asset or a company. A valuation has two components:

- **Projected cash-flows**
- **Interest rate**: determined by risk (uncertainty)

Engineers often need project valuations to make decisions, e.g. lease or buy, replace or keep current equipment, go or no-go on a project.

Why Proper Valuations Are Important

Auger and Guzman (2010) looked at 51 copper-mining investments (1957–1999):

- Half of the projects were invested in at a sub-optimal time
- 36 projects should have been designed with a 40% larger or smaller capacity
- Sub-optimal decisions led to a total NPV **49% lower** than optimal

Conclusion: use better valuation methods (Real Options Analysis).

Risk versus Reward

You have $10,000 to buy a car after graduation (2 years from now). Where do you put it?

1. Bank account
2. Government bond
3. Corporate bond
4. A stock
5. A startup
6. A loan to an acquaintance

Going down this list, the payoff gets more uncertain, so you would demand a higher expected return.

Definition - Financial Risk

The **uncertainty in a future payoff**.

Example Two-State Future

Only two future states are possible.

- Asset A pays $100 in either state
- Asset B pays $120 in state 1 and $70 in state 2

A is risk-free, so we discount it at the risk-free rate. B is risky: what rate should we use? (Answered

by replication in Topic 4.2.)

Definition - Stock Market ("The Market")

The aggregation of buyers and sellers of **stocks** (shares), which represent ownership claims on businesses. Only stocks of "public" companies are traded, usually through brokerages and electronic trading platforms.

Project or Company Returns

With prices P_{t_j} observed every Δt, the (continuously compounded) return over each period is

$$P_{t_j} = P_{t_{j-1}} e^{r_{t_j}\Delta t} \iff r_{t_j} = \frac{1}{\Delta t} \ln \frac{P_{t_j}}{P_{t_{j-1}}}$$

P_{t_j} : price at time t_j
r_{t_j} : return over the period ending at t_j
Δt : time between observations

Stack the returns of the i-th stock into a vector $\vec{R}_i = [r_{t_1}, r_{t_2}, \ldots, r_{t_m}]^T$ and define

$$\sigma_i^2 = \mathrm{Var}(\vec{R}_i), \qquad \textbf{Volatility} = \sigma_i$$

Example Returns from Prices

t	0	1	2	3	4
P	20	21	22	20	21
r_t $(\Delta t = 1)$	–	0.0488	0.0465	−0.0953	0.0488

Mean return $\approx 1.22\%$ per period, volatility $\sigma \approx 0.0717$ (sample std. dev.).

Modern Portfolio Theory (MPT)

- Introduced by Harry Markowitz (1952)
- An investment's risk and return should not be viewed alone, but by how it affects the **overall portfolio's** risk and return
- MPT gives a mathematical framework for building a portfolio that **maximizes expected return for a given level of risk**

Invest a fraction x_i of wealth in the i-th asset:

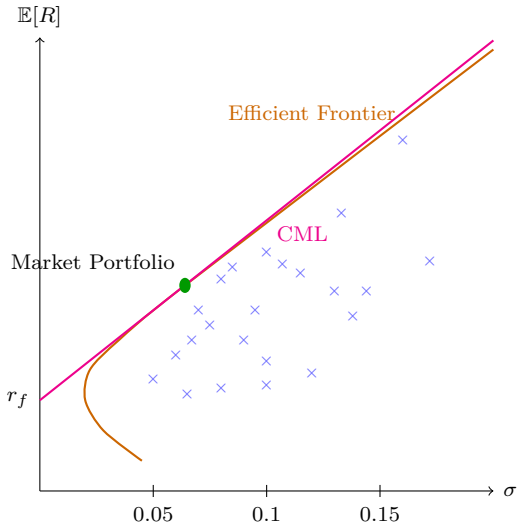
$$\vec{X} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1n} \\ \vdots & \ddots & \vdots \\ \sigma_{n1} & \cdots & \sigma_{nn} \end{bmatrix}, \qquad \sigma_{ii} = \mathrm{Var}(\vec{R}_i),\ \sigma_{ij} = \sigma_{ji} = \mathrm{Cov}(\vec{R}_i, \vec{R}_j)$$

$$\mathbb{E}[R_p] = x_1\mathbb{E}[R_1] + \cdots + x_n\mathbb{E}[R_n], \qquad \sigma_p = \sqrt{\vec{X}^T \Sigma \vec{X}}$$

x_i : fraction of wealth in asset i
$\mathbb{E}[R_i]$: expected return of asset i
Σ : covariance matrix of the asset returns
$\mathbb{E}[R_p]$, σ_p : portfolio expected return and volatility

Optimize: maximize $\mathbb{E}[R_p]$ for a given σ_p.

Efficient Frontier and Capital Market Line



- **Efficient frontier**: the best expected return achievable for each volatility using portfolios of risky assets
- **Capital market line (CML)**: the line from r_f tangent to the frontier. The tangent point is the **market portfolio (MP)**
- Mixing the MP with lending/borrowing at r_f reaches any point on the CML, which **beats the frontier** everywhere else

Example Leveraging Up the CML

$r_f = 3\%$, $\mathbb{E}[R_{MP}] = 7\%$, $\sigma_{MP} = 0.06$. Invest \$100 in the MP, then borrow another \$50 at r_f and invest it in the MP too.
You hold $1.5\times$ your wealth in the MP:

$$\sigma_p = 1.5\,\sigma_{MP} = 0.09, \qquad \mathbb{E}[R_p] = 1.5(7\% - 3\%) + 3\% = 9\%$$

Systematic vs. Idiosyncratic Risk

Regress a company's return on the market's:

$$R_{C,t} = \alpha_C + \beta_C' R_{MP,t} + \varepsilon_{C,t}$$

- Volatility in $R_{MP,t}$ is **systematic** ("market") risk
- Volatility in $\varepsilon_{C,t}$ is **idiosyncratic** ("firm-specific") risk
- Investors can "diversify away" idiosyncratic risk with portfolios, so there is **no reward** for bearing it. Investors are only rewarded for **market risk**, measured by β

CAPM Assumptions

The **Capital Asset Pricing Model (CAPM)** builds on MPT and the CML to give the expected return of any single asset from its market risk (β). It rests on these assumptions:

- Correlations between assets are fixed and constant forever
- All investors maximize economic utility (make as much money as possible) and are rational and risk-averse
- All investors have the same information at the same time, and their probability beliefs match the true distribution of returns
- No taxes or transaction costs
- All investors are **price takers** (their actions do not influence prices)
- Anyone can lend and borrow unlimited amounts at the risk-free rate
- All securities can be divided into parcels of any size

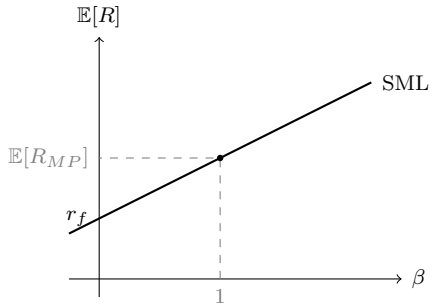
Definition - CAPM

The expected return of the i-th asset is

$$\mathbb{E}[R_i] = r_f + \beta_i(\mathbb{E}[R_{MP}] - r_f), \qquad \beta_i = \frac{\sigma_{i,MP}}{\sigma^2_{MP}} = \frac{\rho_{i,MP}\,\sigma_i}{\sigma_{MP}}$$

r_f : risk-free rate
β_i : sensitivity of asset i to the market (its systematic risk)
$\mathbb{E}[R_{MP}]$: expected return of the market portfolio
$\sigma_i,\ \sigma_{MP}$: volatility of asset i and of the market
$\sigma_{i,MP},\ \rho_{i,MP}$: covariance and correlation of asset i with the market

$\mathbb{E}[R_{MP}] - r_f$ is the **market risk premium**. The MP is often estimated with a stock index.



Security market line: intercept r_f, slope $\mathbb{E}[R_{MP}] - r_f$

Example Estimating Stock Return

Market: $r_f = 2\%$, $\mathbb{E}[R_{MP}] = 8\%$, $\sigma_{MP} = 10\%$. Company X: correlation to the market 0.6, volatility 0.15. Estimate X's expected return.

$$\beta_X = \frac{0.6 \times 0.15}{0.10} = 0.9$$

$$\mathbb{E}[R_X] = 2\% + 0.9(8\% - 2\%) = 7.4\%$$

(The handwritten slide used 10% for the market return, giving 9.2%. With the stated 8% it is 7.4%.)

Example Estimating Future Stock Price

A company with $\beta = 0.9$ trades at \$170/share. $r_f = 2\%$, market rate = 7%. What is the expected share price in 5 years if it pays no dividends?

$$\mathbb{E}[R_C] = 2\% + 0.9(7\% - 2\%) = 6.5\%$$

$$P_5 = P_0 e^{\mathbb{E}[R_C]\, t} = 170 e^{0.065 \times 5} = \$235$$

Example Reflection: Stock with Dividends

A stock trades at \$10 and pays quarterly dividends of \$0.10/share. $\mathbb{E}[R_{MP}] = 10\%$, $r_f = 3\%$, $\beta = 1.2$. What is the expected price 3 years from now?

$$\mathbb{E}[R_C] = 3\% + 1.2(10\% - 3\%) = 11.4\%/\text{yr}$$

Dividends are quarterly, so convert: $(1 + i_q)^4 = 1.114 \Rightarrow i_q = 2.736\%$. Today's price is the PV of 12 dividends plus the price at year 3:

$$10 = 0.10(P/A, 2.736\%, 12) + P_3(P/F, 11.4\%, 3) \;\Rightarrow\; P_3 = \$12.43$$

4.2 - Arbitrage

Definition - Arbitrage

- **In financial markets**: taking advantage of a price difference between two or more markets
- **Academic use**: an opportunity to achieve a **risk-free** gain at a rate **greater than the risk-free rate**
- **Statistical arbitrage**: a gain at a higher rate than one should get for a given level of risk

Fundamental to finance theory (and to valuation) is the **no-arbitrage assumption**: "no free lunch".

Futures and Forwards

- Contracts used to **hedge** against risks or to **speculate**

- Examples of **derivative** assets: their value derives from an underlying asset
- Both lock in a specific price for a certain asset

Definition - Forward Contract

An obligation to buy or sell a certain asset at a specified price (the **forward price**) at a specified time (the **maturity** or expiration date). Typically **not traded on exchanges**; both buyer and seller are **obligated** to fulfill their end at maturity.

Definition - Futures Contract

The same as a forward, except futures are **settled daily** (so they can be bought or sold at any time) and are traded on a **standardized exchange**.

Futures	Forwards
Settled daily	Settled at maturity
Standardized	Not standardized
Low risk of non-fulfilment (regulated)	Low regulation/oversight on settlement
Traded on public exchanges	Private contract between two parties

E.g. the WTI oil futures curve: the price today for oil delivered in month 1, 2, ... (in March 2020 the curve rose steeply from \$25 to \$45).

Example Corn Arbitrage

You can buy a bushel of corn for \$10 today and enter a forward contract to deliver it for \$12 one year from now (storage is free). You can also borrow at 10%. What would you do?
Borrow $\$10x$ and buy x bushels today: net cash-flow \$0. In one year, sell the x bushels for $\$12x$ and repay the $\$11x$ loan: a risk-free profit of $\$1x$.
As $x \to \infty$ the profit is infinite. In reality, demand for corn today pushes the price up until the arbitrage disappears. The no-arbitrage price is

$$\frac{\$12}{1.10} = \$10.91$$

Example Interest Rate Forward

Given a yield curve $r(t)$, find the forward rate $r_{1,2}$ that one could lock in today for borrowing/lending from t_1 to t_2.
Invest P_0 today. Directly to t_2: $F_2 = P_0 e^{r_{0,2} t_2}$. Or to t_1 then roll at the forward rate: $F_2' = P_0 e^{r_{0,1} t_1} e^{r_{1,2}(t_2 - t_1)}$. No arbitrage requires $F_2' = F_2$:

$$r_{1,2} = \frac{r_{0,2} t_2 - r_{0,1} t_1}{t_2 - t_1}$$

$r_{0,t}$: continuously compounded rate from now to t
$r_{1,2}$: forward rate from t_1 to t_2

Example Foreign Exchange Forward

Canadian risk-free rate 3%, US risk-free rate 2.5%, current FX = 0.9 US$/C$. What is the fair one-year forward FX rate?

Two risk-free ways to turn C$1 today into money in one year must agree:

- Keep it in C$: $1 \times 1.03 =$ C$1.03
- Convert to US$0.90, invest: $0.90 \times 1.025 =$ US$0.9225

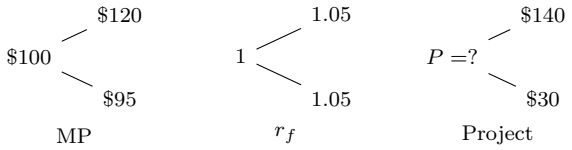
$$FX_1 = \frac{\text{US\$0.9225}}{\text{C\$1.03}} = 0.896 \text{ US\$/C\$}$$

Using Replication to Price an Asset

Build a portfolio of assets we can already price that has the **same payoff in every state**. By no-arbitrage, the asset's price equals the portfolio's price.

Example Pricing a Project by Replication

A project pays $140 if the market goes up and $30 if it goes down. The risk-free rate is 5% for the period; the MP is $100 today and will be worth $120 (up) or $95 (down). What is a fair price for the project?



Hold a units of the MP and b units of the risk-free asset:

$$\text{Up: } 120a + 1.05b = 140$$
$$\text{Down: } 95a + 1.05b = 30$$

$\Rightarrow a = 4.4$, $b = -369.52$ (borrow).

$$P = 100(4.4) - 1(369.52) = \$70.48$$

Note that we did **not** use the probabilities!

Example Expected Returns and the Project's Beta

Same setup; the probability the market goes up is 60%.

$$\mathbb{E}[R_{MP}] = \frac{120(0.6) + 95(0.4)}{100} - 1 = 10\%$$

$$\mathbb{E}[R_{proj}] = \frac{140(0.6) + 30(0.4)}{70.48} - 1 = 36.2\%$$

Back out the project's beta from CAPM:

$$36.2\% = 5\% + \beta_{proj}(10\% - 5\%) \;\Rightarrow\; \beta_{proj} = 6.24$$

The project swings far more than the market, so it is heavily "leveraged" market risk.

Example Payoffs Independent of the Market

a) What if the project paid \$140 regardless of what the market did?
It is risk-free: $P = 140/1.05 = \$133$.

b) What if the payoff had a 50% chance of being \$150 and 50% of being \$130 (independent of the market)?
$\mathbb{E}[\text{payoff}] = 0.5(150) + 0.5(130) = \$140$. The risk is purely idiosyncratic (diversifiable), so it carries no premium: $P = 140/1.05 = \$133$.

Example Systematic and Idiosyncratic Risk

A wealthy (risk-neutral) investor can invest in a product development project. There is a 60% chance the technology works very well (independent of the market). Payoffs:

	Tech works (60%)	Doesn't (40%)
Market up	\$80k	\$50k
Market down	\$55k	\$20k

The MP is \$45 today, worth \$60 (up) or \$40 (down); $r_f = 5\%$. An initial R&D investment of \$20k is required. What is the maximum price for the project?
The technology risk is idiosyncratic, so replace it by its expectation in each market state:

$$\mathbb{E}[P_u] = 0.6(80k) + 0.4(50k) = 68k, \qquad \mathbb{E}[P_d] = 0.6(55k) + 0.4(20k) = 41k$$

Replicate the market risk:

$$60a + 1.05b = 68k$$
$$40a + 1.05b = 41k$$

$\Rightarrow a = 1.35k$, $b = -12.38k$.

$$P_0 = 1.35k(45) - 12.38k = \$48.37k$$

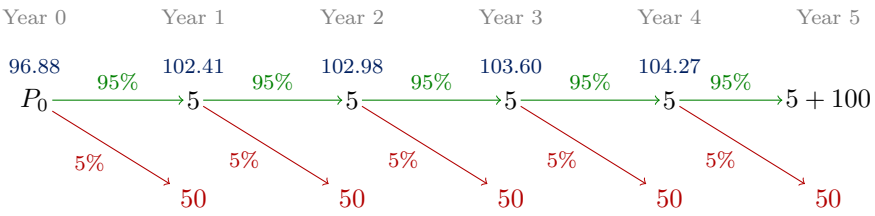
Since \$20k of R&D is also needed, buy the project for **\$28.37k or less**.

Example Bond: Accounting for Default

A company issues a 5-year bond (face \$100) with a 5% annual coupon. The **risk-neutral probability** of default in any given year is 5%, and on default only half of the principal (\$50) is paid out. The risk-free rate is 3%.

a) What is the probability of default over the 5 years?
$1 - 0.95^5 = 22.6\%$

b) What should the price of the bond be?
Work backwards through the tree. Each year the bond either survives (95%, pays the coupon plus the value going forward) or defaults (5%, pays \$50). Discount each step at r_f since these are risk-neutral probabilities:



Survive (green): coupon \$5, plus the \$100 face at year 5. Default (red): \$50 and the bond ends. Blue: bond value at each surviving node, $P_{up}^{(k)}$ (including that year's coupon)

$$P_{up}^{(4)} = 5 + \frac{0.95(105) + 0.05(50)}{1.03} = 104.27$$
$$P_{up}^{(3)} = 5 + \frac{0.95(104.27) + 0.05(50)}{1.03} = 103.60$$
$$P_{up}^{(2)} = 5 + \frac{0.95(103.60) + 0.05(50)}{1.03} = 102.98$$
$$P_{up}^{(1)} = 5 + \frac{0.95(102.98) + 0.05(50)}{1.03} = 102.41$$
$$P_0 = \frac{0.95(102.41) + 0.05(50)}{1.03} = \$96.88$$

c) What is the yield?

$$96.88 = 5(P/A, y, 5) + 100(P/F, y, 5) \;\Rightarrow\; y = 5.735\%$$

The yield is above the 3% risk-free rate (and the 5% coupon) to compensate for default risk.

Summary

1. Risk is the uncertainty in a future payoff; investors demand a higher expected return for bearing it
2. Only **systematic** (market) risk is rewarded. CAPM: $\mathbb{E}[R_i] = r_f + \beta_i(\mathbb{E}[R_{MP}] - r_f)$
3. **No arbitrage**: two ways to get the same payoff must cost the same. This prices forwards (corn, interest rates, FX)
4. **Replication**: match an asset's payoffs with the MP and the risk-free asset; the price does not need the real-world probabilities

Week 5 - Comparison Methods

5.1 - Project Relationships and Alternatives

Definition - Investment ("Project")

An exchange of resources now for expected benefits in the future.

Objective: evaluate and compare projects when there are multiple solutions or opportunities. It is important to consider how these projects are **related** to each other:

- **Independent**
- **Mutually exclusive**
- **Related but not mutually exclusive**

Definition - Independent

The expected costs and benefits of each project **do not depend** on whether or not the other one is chosen.

Example Home Improvement Projects

Paint a room, build a fence, replace the toilet, buy patio furniture, ...
Each one is a separate yes/no decision.

Definition - Mutually Exclusive

When one project is chosen, **all the other ones are excluded**.

Example Buying a Car

You only need one car: buy a Honda Civic, a Toyota Camry, a Volkswagen Jetta, or a Lexus RX, ...
Picking one excludes the rest.

Definition - Related but Not Mutually Exclusive

- **Not mutually exclusive**: you can select more than one
- **Related**: selecting one may affect the selection of another option

A common cause is a **budget limitation**.

Example Reflection: Budget Limits

Project	Budget
1. Paint room	$5000
2. Build a fence	$7000
3. Replace toilet	$6000
4. Buy patio furniture	$3000

How would you describe the relationship between these projects if your total budget is...

a) $1,000,000?
Independent: all four together cost only $21,000, so the budget never binds.

b) $7,000?
Mutually exclusive: the two cheapest together cost $5000 + $3000 = $8000 > $7000, so you can afford at most one.

c) $10,000?
Related but not mutually exclusive: some pairs fit (e.g. 1 & 4), but choosing one project limits which others you can add.

How to Decide on Projects

- **Independent** → consider each option **separately**
- **Mutually exclusive** → find the **best** option
- **Related but not mutually exclusive** → form **mutually exclusive alternatives** (combinations), then find the **best** one

Example Mutually Exclusive Combinations

Same four projects with a limited budget of $10,000. What are the mutually exclusive combinations? List every combination that fits within the budget (including doing nothing):

Alternative	Projects	Cost
0	Do nothing	$0
1	1	$5000
2	2	$7000
3	3	$6000
4	4	$3000
5	1 + 4	$8000
6	2 + 4	$10,000
7	3 + 4	$9000

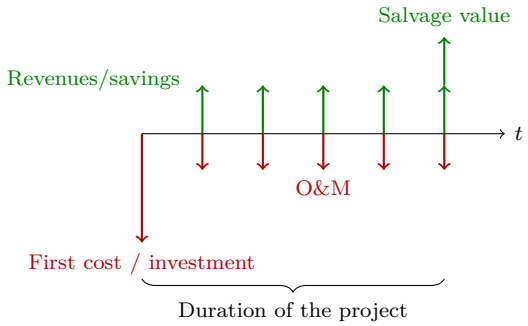
1 + 2 ($12,000), 1 + 3 ($11,000), 2 + 3 ($13,000) and anything larger exceed the budget. Exactly one of these 8 alternatives is chosen, so they are mutually exclusive.

Summary

1. Multiple projects can be **independent**, **mutually exclusive**, or **related but not mutually exclusive**
2. Related projects are handled by forming mutually exclusive combinations

5.2 - Present Worth and Annual Worth

Project Cash-flows



These are discounted using some rate. Which one?

Definition - MARR

The **Minimum Acceptable Rate of Return**: based on risk and/or the expected rate of return from the **best alternative**. Also known as the **hurdle rate**: projects that earn **less than MARR** are not acceptable.

Remark. Assume money can **always** be invested at MARR. This means that **doing nothing** is earning at MARR.

Definition - Present Worth (PW)

Also called **Net Present Value (NPV)**: the present value of benefits − costs, discounted at MARR:

$$\text{PW} = \sum_{t=0}^{N} \frac{F_t}{(1+\text{MARR})^t}$$

F_t : net cash-flow (benefits − costs) at time t
MARR : minimum acceptable rate of return
N : project life (periods)

For **independent** projects:

- PW $> 0 \rightarrow$ acceptable
- PW $< 0 \rightarrow$ unacceptable

Remark. The PW of a project is "the amount by which this project is beating the best alternative, expressed in today's value."

Example Calculating PW

You plan to buy a Honda Civic with a life of 10 years; MARR is 5%. The annual benefits are worth \$4000 and the annual costs \$1000. The first cost is \$20,000 and the salvage value after 10 years is \$4000. What is the PW?
Net annual benefit $= 4000 - 1000 = \$3000$:

$$\begin{aligned} \text{PW} &= -20{,}000 + 3000(P/A, 5\%, 10) + 4000(P/F, 5\%, 10) \\ &= -20{,}000 + 3000(7.7217) + 4000(0.61391) = \$5621 \end{aligned}$$

PW > 0, so buying the car is acceptable.

Definition - Annual Worth (AW)

The **equivalent annuity** of PW:

$$\text{AW} = \text{PW}(A/P, \text{MARR}, N)$$

PW : present worth of the project
MARR : minimum acceptable rate of return
N : project life (periods)

For **independent** projects:

- AW $> 0 \rightarrow$ acceptable
- AW $< 0 \rightarrow$ unacceptable

Example Annual Worth of Car

Honda Civic: FC \$20,000, annual benefits − costs \$3000, life 10 years, salvage \$4000, MARR 5%.
Convert each cash-flow to an annuity directly:

$$\begin{aligned} \text{AW} &= -20{,}000(A/P, 5\%, 10) + 3000 + 4000(A/F, 5\%, 10) \\ &= -20{,}000(0.12950) + 3000 + 4000(0.07950) = \$728 \end{aligned}$$

Check: $5621(A/P, 5\%, 10) = 5621(0.12950) = \$728$.

Summary

1. MARR is the rate of return of the best alternative, used as the "hurdle" rate
2. PW is the present value of all costs/benefits discounted at MARR; AW is its equivalent annuity
3. Independent projects: accept if PW > 0 (equivalently AW > 0)

5.3 - Evaluating Mutually Exclusive Projects (PW/AW)

Mutually Exclusive Projects

We can only choose one option, so we must choose the **best**. Steps to evaluate projects with PW:

1. Define the **time horizon**
2. Develop the cash-flows for each alternative
3. Calculate the PW using MARR
4. Compare the PWs and pick the best

Remark. "Do nothing" is always an alternative when no better choice is available (i.e. if all options have PW < 0).

Example **Mutually Exclusive Projects with Same Lives**

	Honda Civic	Tesla Model S
FC	\$20,000	\$100,000
Annual benefits − cost	\$3000	\$12,000
Life	10 years	10 years
Salvage value	\$4000	\$14,000
MARR	5%	5%

Both lives are 10 years, so compare PWs directly:

$$\text{PW}_H = -20{,}000 + 3000(P/A, 5\%, 10) + 4000(P/F, 5\%, 10) = \$5621$$
$$\text{PW}_T = -100{,}000 + 12{,}000(7.7217) + 14{,}000(0.61391) = \$1256$$

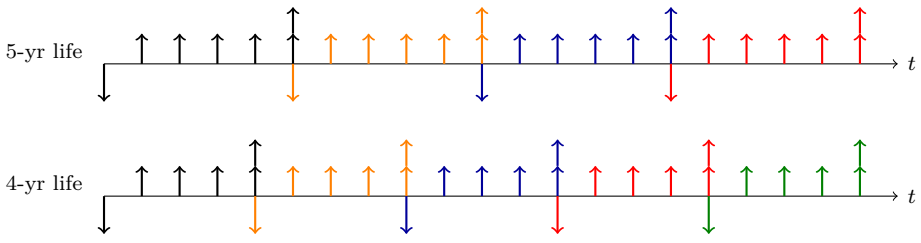
Both beat doing nothing, but $\text{PW}_H > \text{PW}_T$: **buy the Honda Civic**.

PW Comparison for Different Lives

What if the lives (and so the time horizons) differ? Consider buying a toothbrush: one costs \$5 and lasts 2 years, the other costs \$4 and lasts 1 year.
Comparing \$5 against \$4 directly is unfair: the \$4 one only covers half the time. We need a common time horizon.

Definition - Repeated Lives

> Assume each project **repeats itself** (e.g. you always need a car/toothbrush). Compare over the **least common multiple (LCM)** of the individual lives.



Lives of 5 and 4 years both repeat to fill the LCM of 20 years

Example PW (Repeated Lives)

	Honda Civic	Lexus RX
FC	\$20,000	\$40,000
Annual benefits − cost	\$3000	\$8000
Life	10 years	**5 years**
Salvage value	\$4000	\$13,000
MARR	5%	5%

LCM of 10 and 5 is 10 years, so the Lexus is bought twice. At $t = 5$ it is sold for \$13,000 and replaced for \$40,000, a net −\$27,000:

$$\begin{aligned} \text{PW}_L &= -40{,}000 + 8000(P/A, 5\%, 10) + 13{,}000(P/F, 5\%, 10) - 27{,}000(P/F, 5\%, 5) \\ &= -40{,}000 + 8000(7.7217) + 13{,}000(0.61391) - 27{,}000(0.78353) \\ &= \$8600 \end{aligned}$$

$\text{PW}_H = \$5621$ (from before). $\text{PW}_L > \text{PW}_H$: **buy the Lexus RX**.

AW of Repeated Lives

We can also compare options using AW (the equivalent annuity of PW). A repeated project has the **same** AW in every cycle, so the AW over one life equals the AW over the LCM.

Remark. Comparing AW is equivalent to comparing PW of repeated projects, and only needs **one life** of each project.

Example AW (Repeated Lives)

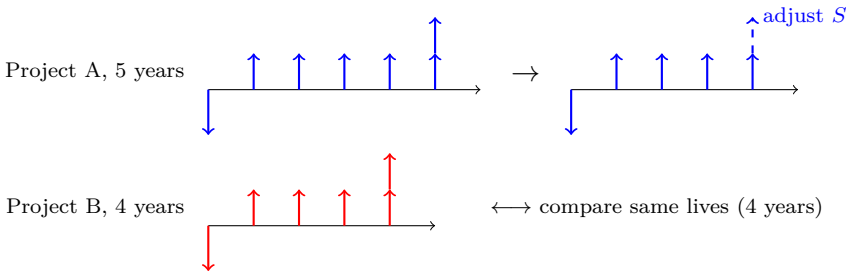
Same Honda Civic and Lexus RX.

$$\begin{aligned} \text{AW}_H &= -20{,}000(A/P, 5\%, 10) + 3000 + 4000(A/F, 5\%, 10) = \$728 \\ \text{AW}_L &= -40{,}000(A/P, 5\%, 5) + 8000 + 13{,}000(A/F, 5\%, 5) \\ &= -40{,}000(0.23097) + 8000 + 13{,}000(0.18097) = \$1114 \end{aligned}$$

$\text{AW}_L > \text{AW}_H$: **buy the Lexus RX**, the same decision as PW. Check: $8600(A/P, 5\%, 10) = \$1114$.

Definition - Study Period

Adopt a **specified time period** for comparison (e.g. when projects will not be repeated), and **adjust the salvage value** of any project whose life extends past the study period.



Example Study Period

Use a 5-year study period. If the Honda Civic is sold after 5 years (instead of 10), its salvage value is \$7000 (instead of \$4000).

	Honda Civic	Lexus RX
FC	\$20,000	\$40,000
Annual benefits − cost	\$3000	\$8000
Life	**5** (was 10) years	5 years
Salvage value	**\$7000** (was \$4000)	\$13,000
MARR	5%	5%

$$\begin{aligned} \text{PW}_H &= -20{,}000 + 3000(P/A, 5\%, 5) + 7000(P/F, 5\%, 5) \\ &= -20{,}000 + 3000(4.3295) + 7000(0.78353) = -\$1527 \\ \text{PW}_L &= -40{,}000 + 8000(4.3295) + 13{,}000(0.78353) = \$4822 \end{aligned}$$

$\text{PW}_L > \text{PW}_H$: **buy the Lexus RX**. Note the Honda alone would be rejected ($\text{PW}_H < 0$) over a 5-year horizon.

Summary

Choosing the best option among mutually exclusive projects:

1. **Same time horizons**: compare PW
2. **Different time horizons, repeated projects**: PW of repeated lives (LCM) or AW
3. **Fixed time horizon**: PW over a defined study period (adjust salvage values)

Week 6 - Comparison Methods (Part 2)

6.1 - Internal Rate of Return

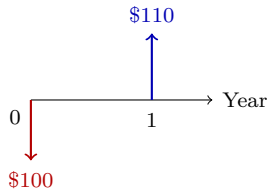
Rate of Return Analysis

Cash-flow equivalence methods (PW, AW) require us to know the discount rate **before** we start. **Rate of return analysis** instead *starts* by asking what *is* the rate of return on an investment. Compare:

- "The present worth of this project is \$32,000"
- "The annual worth of this project is \$2,800"
- "The project will produce a 23% rate of return"

Example Rate of Return

If \$100 is invested today in a project that returns \$110 next year, what is the rate of return?



$$P = F(P/F, i^*, 1) \;\Rightarrow\; 100 = \frac{110}{1+i^*} \;\Rightarrow\; i^* = 10\%$$

A 10% rate makes the PW of benefits equal to the PW of costs/investments.

Definition - Internal Rate of Return (IRR)

The discount rate at which the **present worth of a project is equal to 0**:

$$\mathrm{PW}(i^*) = \sum_{t=0}^{N} \frac{F_t}{(1+i^*)^t} = 0$$

i^* : internal rate of return
F_t : net cash-flow at time t
N : project life (periods)

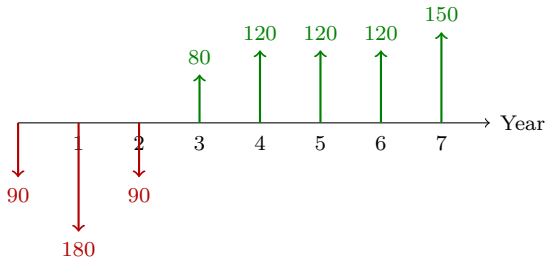
Equivalently, the discount rate that makes

$$\mathrm{PW}(\text{benefits}) = \mathrm{PW}(\text{costs}) \quad \text{or} \quad \mathrm{AW}(\text{benefits}) = \mathrm{AW}(\text{costs})$$

More profitable projects → **higher IRR**.

Example IRR of an Investment Project

Find the IRR of the following project (cash-flows in \$K).



$$\mathrm{PW}(i) = -90 - \frac{180}{1+i} - \frac{90}{(1+i)^2} + \frac{80}{(1+i)^3} + \frac{120}{(1+i)^4} + \frac{120}{(1+i)^5} + \frac{120}{(1+i)^6} + \frac{150}{(1+i)^7}$$

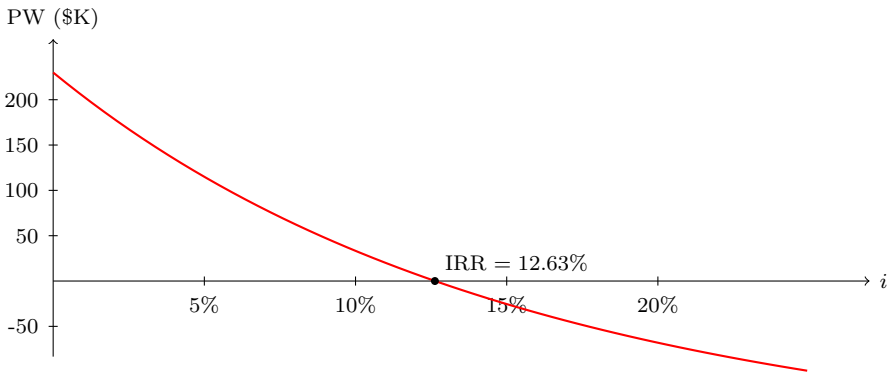
There is no closed form, so estimate. Evaluate PW at a few rates (with $(P/F, i, N)$ factors):

i	0%	5%	10%	15%	20%
PW (\$K)	230.0	114.9	33.3	−25.4	−68.1

PW changes sign between 10% and 15%. Linear interpolation:

$$i^* \approx 10\% + 5\% \cdot \frac{33.3}{33.3 - (-25.4)} = 12.8\%$$

The exact value (Excel) is $i^* = \mathbf{12.63\%}$. Interpolation slightly overestimates because the curve is convex.



For simple investments (costs earlier than benefits), higher i gives lower PW

Using MS Excel to Find IRR

- **NPV**: finds the present worth, `=NPV(i, cash flows)`. NB: it **starts with year 1**! Add the year-0 cash-flow separately
- **IRR**: finds the IRR, `=IRR(cash flows, guess)`. Include the year-0 cash-flow

	A	B	C
1	Year	Cash flow	
2–9	0, 1, . . . , 7	−90, −180, . . . , 150	
11	IRR	12.63%	`=IRR(B2:B9,12%)`
13	PW	$0.0	`=B2+NPV(12.63%,B3:B9)`
14	PW	$114.94	`=B2+NPV(5%,B3:B9)`

Example IRR: Analytical Solution

$5,000 is invested today in a project that returns $5,500 in 2 years. What is the IRR?
Only two cash-flows, so solve directly:

$$5000 = 5500(P/F, i^*, 2) = \frac{5500}{(1+i^*)^2} \Rightarrow i^* = \sqrt{1.1} - 1 = 4.88\%$$

Definition - Simple Investment

A project where **all negative cash-flows occur before all positive cash-flows**. For simple investments, evaluating an **independent** project with IRR:

- IRR > MARR → worthwhile (PW > 0)
- IRR < MARR → not worthwhile (PW < 0)

In the investment project above: at MARR = 5% < 12.63%, PW = $114.94K > 0. The PW curve is decreasing, so IRR > MARR $\iff$ PW(MARR) > 0.

Limitation of IRR: Not a Simple Investment

- When cash-flows change sign (+ → − or − → +) **more than once**, calculating IRR becomes complicated
- There can be **multiple** discount rates that make PW = 0

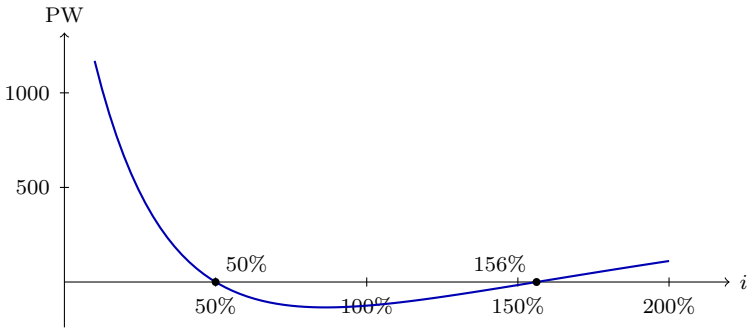
Example Not a Simple Investment

Year 0: +1000; Year 1: −2500; Year 2: −2500; Year 3: +6000.
The signs go + → − → + (two changes). Set $x = 1 + i$:

$$1000x^3 - 2500x^2 - 2500x + 6000 = 0 \Rightarrow (x - 1.5)(x^2 - x - 4) = 0$$

$$x = 1.5 \text{ or } x = \tfrac{1+\sqrt{17}}{2} = 2.562 \Rightarrow i^* = 50\% \text{ and } i^* = 156\%$$

(The third root $x < 0$ is not a meaningful rate.) Which one is "the" IRR? Neither is a useful measure of profitability.



Summary

1. **IRR**: the discount rate that makes PW = 0
2. For simple investments, IRR is a useful measure of profitability. For independent projects, compare IRR to MARR
3. IRR is useful for **simple investments only**: if cash-flows change sign more than once, there can be multiple IRRs

6.2 - Comparing Mutually Exclusive Projects Using IRR

Comparing Mutually Exclusive Options

IRR is an **efficiency** (rate) of earning money, but it can be misleading when comparing mutually exclusive options.

Example Reflection: Higher IRR?

Project A has an IRR of 50%; project B has an IRR of 40%. The two projects are mutually exclusive. Which option is better?
We can't tell from IRR alone: it says nothing about how much money is invested.

Example Reflection: Lending to George

You have \$20 and your friend George wants to borrow some money. You can either:

A. Lend him \$10 and receive \$15 in return (1 year); or
B. Lend him \$20 and receive \$28 in return (1 year)

MARR is 10%. Which option should you choose?

$$\text{A: IRR} = 15/10 - 1 = 50\%, \qquad \text{PW} = -10 + 15/1.1 = \$3.64$$
$$\text{B: IRR} = 28/20 - 1 = 40\%, \qquad \text{PW} = -20 + 28/1.1 = \$5.45$$

A has the higher IRR, but B leaves you better off. With A, the other \$10 only earns MARR. Look at the **increment** B − A: lend an extra \$10 now, get an extra \$13 back:

$$\text{IRR}_{B-A} = 13/10 - 1 = 30\% > 10\% \Rightarrow \textbf{choose B}$$

Remark. A higher IRR does not mean a better project: IRR is a rate of earning money, and just looking at it **ignores the size of the initial investment**.

Definition - Incremental IRR

The IRR of the **difference** (the "increment") between two mutually exclusive alternatives. Steps in incremental analysis:

1. Order alternatives in **increasing order of first cost**
2. Start with the "do nothing" alternative as the current best
3. One by one, evaluate each alternative against the **best alternative found so far**: compute the IRR of (challenger − current best)
   - $\text{IRR}_\Delta > \text{MARR} \to$ the extra investment is worthwhile; the challenger becomes the current best
   - $\text{IRR}_\Delta < \text{MARR} \to$ keep the current best
4. Repeat step 3 until all alternatives are evaluated

Example IRR of Mutually Exclusive Projects

ABC company must purchase a new lathe. It is considering 4 lathes with a 10-year lifetime and no salvage value. Which should be taken with a MARR of 15%?
Order by increasing first cost:

Lathe	First cost	Annual savings	IRR (alone)
1	100,000	25,000	21.4%
2	150,000	34,000	18.5%
3	200,000	46,000	18.9%
4	255,000	55,000	17.1%

Each increment is a first cost ΔP followed by 10 annual savings ΔA, so solve $(P/A, i^*, 10) = \Delta P/\Delta A$:

Comparison	ΔP	ΔA	IRR_Δ	Decision
1 vs. do nothing	100,000	25,000	$21.4\% > 15\%$	take 1
2 vs. 1	50,000	9,000	$12.4\% < 15\%$	keep 1
3 vs. 1	100,000	21,000	$16.4\% > 15\%$	take 3
4 vs. 3	55,000	9,000	$10.1\% < 15\%$	keep 3

Choose lathe 3, even though lathe 1 has the highest IRR. Check with PW at 15%, using $(P/A, 15\%, 10) = 5.0188$:

$$\text{PW}_1 = \$25{,}469, \quad \text{PW}_2 = \$20{,}638, \quad \text{PW}_3 = \$30{,}863, \quad \text{PW}_4 = \$21{,}032$$

Lathe 3 has the highest PW, so the two methods agree.

Summary

1. When comparing **mutually exclusive** options, simply comparing IRRs ignores the size of the initial investment and may lead to the **wrong conclusion**
2. Use **incremental IRR** instead: it gives the same choice as PW

6.3 - Payback Period

Definition - Payback Period

The time it takes for an investment to be **recouped**: sum the revenues/savings until they equal the initial investment.

Definition - Discounted Payback Period

Sum the **PW** of the revenues/savings (discounted at MARR) until they equal the initial investment.

Example Payback Period

Find the payback period for the following cash-flows. First cost = \$100,000.

N	Cash-flow	Cumulative
1	\$10,000	\$10,000
2	\$20,000	\$30,000
3	\$25,000	\$55,000
4	\$30,000	\$85,000
5	\$31,000	\$116,000
6	\$32,000	\$148,000
7	\$34,000	\$182,000

The cumulative total passes \$100,000 during year 5. After year 4, \$15,000 is still owed out of year 5's \$31,000:

$$\text{Payback} = 4 + \frac{15{,}000}{31{,}000} = 4.48 \text{ years}$$

Example Discounted Payback Period

Same cash-flows, MARR = 10%, FC = \$100,000.

N	Cash-flow	$(P/F, 10\%, N)$	PW	Cumulative PW
1	\$10,000	0.90909	\$9,091	\$9,091
2	\$20,000	0.82645	\$16,529	\$25,620
3	\$25,000	0.75131	\$18,783	\$44,403
4	\$30,000	0.68301	\$20,490	\$64,893
5	\$31,000	0.62092	\$19,249	\$84,142
6	\$32,000	0.56447	\$18,063	\$102,205
7	\$34,000	0.51316	\$17,447	\$119,652

The cumulative PW passes \$100,000 during year 6:

$$\text{Discounted payback} = 5 + \frac{100{,}000 - 84{,}142}{18{,}063} = 5.88 \text{ years}$$

Longer than the simple payback, since later dollars are worth less today.

Pros and Cons

- A shorthand answer to the question "**When will I make my money back?**"
- Commonly used because it is **simple to calculate and understand**
- **Not a sound method** in engineering economics, and against its principles: it ignores everything that happens **after** the payback point (and the simple version ignores the time value of money)

Example Reflection: Payback vs. PW

When will payback period give a different preference from AW or PW?
When cash-flows **after** the payback point differ. E.g. at MARR = 10%, project X costs \$100 and returns \$100 in year 1 only; project Y costs \$100 and returns \$50/year for 5 years.

$$\text{X: payback} = 1 \text{ yr}, \quad \text{PW} = -100 + 100(0.90909) = -\$9.09$$
$$\text{Y: payback} = 2 \text{ yr}, \quad \text{PW} = -100 + 50(P/A, 10\%, 5) = -100 + 50(3.7908) = \$89.54$$

Payback prefers X, but X doesn't even earn MARR; PW/AW correctly prefer Y.

Summary

1. **Payback period**: simple concept and easy to calculate
2. **Discounted payback period** accounts for the time value of money
3. Both fail to account for the profitability of the project **after** the investment is recovered